

Banking Loan Performance & Credit Risk Analysis

Project Documentation

1. Project Overview

The **Banking Loan Performance & Credit Risk Analysis** project was developed to analyze lending activity, loan repayment performance, and credit-risk patterns using SQL Server and Power BI.

The objective of the project was to understand:

- Overall loan portfolio performance
- Loan application and funding trends
- Good loans versus bad loans
- Credit risk across loan grades
- Risk associated with different loan terms and purposes
- Borrower characteristics such as income, DTI, and home ownership
- Charged-off loan exposure
- Key factors associated with higher-risk lending segments

The project followed an end-to-end Data Analyst workflow:

Raw Dataset → SQL Server → Data Profiling → Business Analysis → Power Query → DAX → Power BI Dashboard → SQL Validation

2. Tools Used

- **SQL Server / SSMS** – Data profiling, business analysis, and dashboard validation
 - **Power BI** – Dashboard development and visualization
 - **Power Query** – Data cleaning and transformation
 - **DAX** – KPI and analytical measure creation
 - **Excel** – Original dataset format
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3. Dataset Overview

The project used a financial loan dataset containing detailed information about individual loans and borrowers.

Dataset Size

- Total Records: **38,576**
- Unique Loan IDs: **38,576**
- Total Columns: **24**
- Data Period: **2021**

Each row represents one loan application.

Important fields included:

- Loan ID
- Loan Amount
- Loan Status
- Interest Rate
- DTI
- Annual Income
- Grade
- Sub Grade
- Loan Purpose
- Loan Term
- Home Ownership
- Employment Length
- Employment Title
- State
- Issue Date
- Total Payment
- Verification Status

4. Data Profiling Using SQL

Initial SQL profiling was performed to understand the quality, structure, and completeness of the dataset.

The following checks were completed:

- Total row count
- Unique loan count
- Duplicate loan ID check
- Loan-status distribution
- Missing-value analysis
- Issue-date range
- Overall financial KPIs

Loan Status Distribution

- Fully Paid: **32,145**
- Charged Off: **5,333**
- Current: **1,098**

No duplicate loan IDs were identified.

The main missing-value issue was found in:

emp_title

with approximately **1,438 missing values**.

5. Good Loan and Bad Loan Classification

For analytical purposes, loans were grouped into two categories.

Good Loans

Loans with status:

- Fully Paid
- Current

Total Good Loans:

33,243

Good Loan Percentage:

86.18%

Bad Loans

Loans with status:

- Charged Off

Total Bad Loans:

5,333

Bad Loan Percentage:

13.82%

This classification was used throughout the credit-risk dashboard.

6. SQL Business Analysis

Business analysis was performed in SQL before creating the Power BI dashboard.

The main analysis areas included:

Overall Loan Performance

Calculated:

- Total Loan Applications
- Total Funded Amount
- Total Amount Received
- Average Annual Income
- Average Interest Rate
- Average DTI

Key overall values:

- Loan Applications: **38,576**
- Total Funded Amount: approximately **\$435.76M**
- Total Amount Received: approximately **\$473.07M**
- Average Interest Rate: approximately **12.05%**
- Average DTI: approximately **13.33%**

Monthly Loan Trends

Loan applications, funded amount, and amount received were analyzed by issue month to understand lending activity throughout the year.

Credit Risk by Grade

Bad loan rates were analyzed across credit grades.

The analysis showed that risk generally increased as the loan grade weakened.

Approximate bad-loan rates included:

- Grade A: **5.70%**
- Grade B: **11.50%**
- Grade C: **16.02%**
- Grade D: **20.69%**
- Grade E: **24.80%**
- Grade F: **30.25%**
- Grade G: **31.31%**

This demonstrated a clear relationship between credit grade and loan performance.

Loan Risk by Term

Loan performance was compared between:

- 36-month loans
- 60-month loans

Approximate bad-loan rates:

- 36 Months: **10.71%**
- 60 Months: **22.34%**

The 60-month loans showed considerably higher risk.

Additional Analysis

SQL analysis also covered:

- Loan purpose
- Home ownership
- State
- Employment length
- Borrower risk profile

- Interest rate differences by loan status

7. Data Cleaning and Transformation in Power Query

The dataset was loaded into Power BI and transformed using Power Query.

Data Type Validation

Appropriate data types were assigned to:

- Date fields
- Numeric fields
- Percentage fields
- Text fields
- Identifier fields

Missing Employment Titles

Missing emp_title values were handled by assigning:

Not Provided

This preserved the records while clearly identifying unavailable employment information.

Loan Category

A new field called:

Loan Category

was created.

Logic:

- Fully Paid / Current → **Good Loan**
- Charged Off → **Bad Loan**

Date Fields

Supporting date columns were created from issue_date:

- Issue Month
- Issue Month Number

Issue Month Number was used to ensure months were displayed chronologically in Power BI.

8. DAX Measures

Several DAX measures were created to support the dashboards.

Main measures included:

- Total Loan Applications
- Total Funded Amount
- Total Amount Received
- Average Interest Rate
- Average DTI
- Good Loan Applications
- Bad Loan Applications
- Good Loan %
- Bad Loan %
- Good Loan Funded Amount
- Bad Loan Funded Amount
- Good Loan Amount Received
- Bad Loan Amount Received

These measures were used dynamically with dashboard slicers and visual filters.

9. Dashboard 1 – Loan Performance Overview

The first dashboard page was designed to provide an executive overview of lending activity.

KPI Cards

The dashboard included:

- Loan Applications
- Funded Amount
- Amount Received
- Average Interest Rate

- Average DTI

Slicers

Interactive filtering was added for:

- Loan Grade
- Loan Purpose
- Loan Term

Visuals

The page included:

- Monthly Loan Application Trend
- Top Loan Purposes by Applications
- Loan Applications by Grade
- Loan Applications by Term
- Loan Applications by Home Ownership
- Top States by Loan Applications

This page provides a high-level view of lending volume, customer characteristics, and portfolio distribution.

10. Dashboard 2 – Credit Risk Analysis

The second dashboard focused specifically on loan quality and credit risk.

KPI Cards

The page included:

- Good Loans
- Good Loan %
- Bad Loans
- Bad Loan %
- Bad Loan Funded Amount

Slicers

Filtering was provided for:

- Loan Grade
- Loan Term
- Home Ownership

Risk Analysis Visuals

The dashboard included:

- Loan Quality Distribution
- Bad Loan Rate by Grade
- Bad Loan Rate by Loan Term
- Bad Loan Rate by Loan Purpose
- Average Interest Rate by Loan Quality

Charged-Off Loan Details

A detailed table was added for charged-off loans containing:

- Loan ID
- Grade
- Sub Grade
- Purpose
- Term
- Annual Income
- Interest Rate
- DTI
- Loan Amount
- Amount Received

The table was filtered specifically to bad loans to support detailed risk investigation.

11. Dashboard Design and Formatting

A consistent banking-style theme was applied across both dashboards.

Main colors used:

- Navy – Titles and neutral information
- Blue – General loan performance
- Green – Good loan indicators
- Red – Credit risk / bad loans
- Orange – Financial exposure
- White – Dashboard cards
- Light grey-blue – Page background and borders

Consistent formatting was applied to:

- KPI cards
- Borders
- Rounded corners
- Shadows
- Slicers
- Chart titles
- Data labels
- Table headers
- Visual spacing and alignment

12. SQL Dashboard Validation

After completing the dashboards, Power BI results were validated against SQL.

Validation included:

- Total Loan Applications
- Total Funded Amount
- Total Amount Received
- Average Interest Rate
- Average DTI
- Good Loan Count

- Bad Loan Count
- Good Loan %
- Bad Loan %
- Bad Loan Funded Amount
- Monthly loan trends
- Loan applications by purpose
- Grade-level performance
- Loan term analysis
- Home ownership
- State-level loan applications
- Bad loan rate by purpose
- Average interest rate by loan quality

The SQL outputs were compared with Power BI KPI cards, chart values, and tooltips to confirm calculation accuracy.

13. Key Business Insights

The analysis identified several important findings.

Overall loan quality was strong, with approximately **86.18% of loans classified as good loans**, while **13.82% were charged off**.

Credit risk increased considerably as loan grade declined. Lower-grade loans showed much higher bad-loan rates than Grade A and Grade B loans.

60-month loans carried substantially greater risk than 36-month loans, with a bad-loan rate of approximately 22.34% compared with around 10.71%.

Charged-off loans generally carried higher interest rates, indicating that higher-risk borrowers were priced at higher borrowing costs.

Loan purpose also influenced portfolio risk, allowing higher-risk borrowing purposes to be identified and monitored.

The charged-off loan details table provided loan-level visibility into higher-risk accounts and financial exposure.

14. Project Outcome

The project demonstrates an end-to-end Data Analyst workflow covering:

SQL → Data Cleaning → Business Analysis → DAX → Power BI → Validation

The final dashboards allow decision-makers to monitor both overall lending performance and credit risk from a single interactive report.

The project demonstrates practical skills in:

- SQL analysis
- Financial data analysis
- Credit-risk analysis
- Data cleaning
- Power Query
- DAX
- Power BI
- Data visualization
- KPI development
- Dashboard design
- Data validation

15. Dataset Disclaimer

This project uses a publicly available financial loan dataset for **educational and portfolio purposes**.